



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin
Ollscoil Átha Cliath | The University of Dublin

Faculty of Science, Technology, Engineering and Mathematics

School of Computer Science & Statistics

Integrated Computer Science
Joint Honors
Management Science and Information Systems Studies
Year 3 Annual Examinations

Semester 2 2025

Information Management II

24th April 2025

09:30 - 11:30

Prof Yvette Graham

Instructions to Candidates:

Attempt **three** questions in total. **Question 1 is mandatory**. Answer **any two** questions from **Question 2, Question 3 and Question 4**

All questions carry equal marks. Each question is scored out of a total of 25 marks.

Do not begin the exam until instructed to do so by an invigilator.

Materials permitted: None

Q. 1

A relational database is required to keep track of transport ships and their locations for maritime authorities, recording the movement of ships between ports and the current location of a given ship. The database should keep track of the types of ships stored in the database by recording the tonnage and hull type of each ship, which can be any one of the following: flat-bottom, round-bottom, v-shaped, or multi-hulled. The database also needs to record information about the ports that each ship visits, including the country the port is located in, and the continent of that country, in addition to the sea, ocean or lake it is located on. Each ship will have a home port, the port it is ordinarily docked in. Each visit to a port by a ship is recorded as well as the port at which a given ship is docked at any given time (it can alternatively be in transit). The current location of a ship is recorded at 5 minute intervals, and at any point in time, the database is able to retrieve a complete ship movement history for any ship in the database.

- (i) Provide a suitable Entity Relationship schema for the above SHIP_TRACKING database. Make sure to state any assumptions you make.

[10 marks]

- (ii) Map the ER schema into a relational database schema and specify all primary and foreign keys. Again, state any assumptions you make.

[15 marks]

Q. 2

A relational database is required to record airline flight information. Each flight should be identified by a flight number, and consists of one or more flight legs with leg numbers 1, 2, 3, and so on. Each flight leg has scheduled arrival and departure times, airports, and one or more leg instances, one for each date on which the flight travels. Fares are also recorded for each flight. For each flight leg instance, seat reservations are kept, as are the airplane used on the leg and the actual arrival and departure times and airports. An airplane is identified by an airplane id and is of a particular airplane type. A record needs to be kept of which airplane types are allowed to land at which airports, with distinct airports identified by a unique airport code. The relational schema is shown below.

AIRPORT

<u>Airport_code</u>	Name	City	State
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FLIGHT

<u>Flight_number</u>	Airline	Weekdays
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FLIGHT_LEG

<u>Flight_number</u>	<u>Leg_number</u>	Departure_airport_code	Scheduled_departure_time
		Arrival_airport_code	Scheduled_arrival_time

LEG_INSTANCE

<u>Flight_number</u>	<u>Leg_number</u>	<u>Date</u>	Number_of_available_seats	Airplane_id		
			Departure_airport_code	Departure_time	Arrival_airport_code	Arrival_time

FARE

<u>Flight_number</u>	<u>Fare_code</u>	Amount	Restrictions
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AIRPLANE_TYPE

<u>Airplane_type_name</u>	Max_seats	Company
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CAN_LAND

<u>Airplane_type_name</u>	<u>Airport_code</u>
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AIRPLANE

<u>Airplane_id</u>	Total_number_of_seats	Airplane_type
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SEAT_RESERVATION

<u>Flight_number</u>	<u>Leg_number</u>	<u>Date</u>	<u>Seat_number</u>	Customer_name	Customer_phone
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Specify the following queries in relational algebra for the above database.

- (a) For each flight, list the flight number, airline, date, departure airport and arrival airport along with the number of available seats, for each leg.
[5 marks]
- (b) For all the airplane types available in the database, list the airplane type, name of the company, maximum seats available and the airports (or airport codes) where these planes can land.
[5 marks]
- (c) List the name, seat number and phone numbers of all the customers of all flights or flight legs that departed from Houston Intercontinental Airport (airport code 'iah') and arrived in Los Angeles International Airport (airport code 'lax') on '2016-03-16'.
[5 marks]
- (d) List fare information for all the flights run by the airline 'Delta Airlines'.
[5 marks]
- (e) Retrieve the number of available seats on all flights run by Delta Airline, on '2016-04-09'.
[5 marks]

Q 3.

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

The above relational database stores student and course information for a university where information about primary and foreign keys is not explicitly stated. Specify the following queries in SQL on the database schema.

- (i) Create each of the tables in the above schema. Making sure to appropriately apply constraints, include primary and foreign keys.

[4 marks]

- (ii) Retrieve the names of all courses along with the name of the instructor taught during the fall of 2008.
[4 marks]
- (iii) For each section taught by Professor Anderson, retrieve the course number, semester, year, and number of students who took the section.
[4 marks]
- (iv) Retrieve the name and transcript of each junior student (Class = 1) majoring in mathematics (MATH). A transcript includes course name, course number, credit hours, semester, year, and grade for each course completed by the student.
[4 marks]
- (v) Retrieve the names and major departments of all straight-A students (students who have a grade of A in all their courses).
[4 marks]
- (vi) Retrieve the names and major departments of all students who do not have any grade of A in any of their courses.
[5 marks]

Q 4.

- (i) Consider the following two transactions, T1 and T2, and their operations. Analyze the following operations and explain whether T2 operations are updated successfully when T1 is rollback.

[5 marks]

T1	T2
Read(X)	
$X = X - 2$	
Write(X)	
	Read(x)
	$X = X + 3$
	Write(X)
ROLLBACK	
	COMMIT

- (ii) Does strict two-phase locking always guarantee strict schedules? Explain your answer in detail.

[12 marks]

- (iii) Provide a definition for deadlock.

[3 marks]

- (iv) Does cautious waiting avoid deadlock?

[5 marks]